



ELEMENTARY MATH (CALCULUS)



INTEGRALS: INDEFINITE AND DEFINITE INTEGRALS

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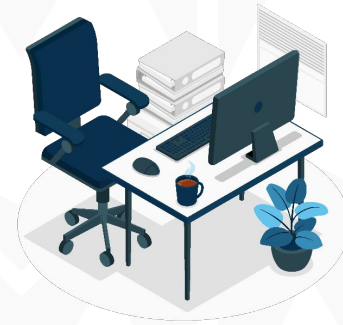
MSc Mathematics



LEARNING OBJECTIVES

Upon completion of this unit, you should be able to:

- Know the difference between indefinite and definite integrals,
- Solve problems involving indefinite integrals.
- Solve problems involving definite integrals.





INTRODUCTION

When we differentiate a function, we see its rate of change. When we integrate a function, we see its accumulated value. In this note, we will explore the meaning of, and difference between indefinite and definite integrals. The indefinite integral of a function is the collection of all its possible antiderivatives. The definite integral of a function over an interval is the total area between the curve and the independent-variable axis in that interval.





INDEFINITE INTEGRALS

Suppose that f is a continuous real-number function of a variable x . The indefinite integral of f is a function F such that

$$F'(x) = f(x)$$

added to the constant of integration c . This is written as follows:

$$\int f(x)dx = F(x) + c.$$





INDEFINITE INTEGRALS

Example 1.

Consider the function $f(x) = x^2$. The derivative is $f'(x) = 2x$. Now think about the following indefinite integral:

$$\int 2x dx.$$

Let's rename the function we're integrating. We can say that $g(x) = 2x$. At first thought, it's tempting to suppose that the antiderivative of g is the function G such that $G(x) = x^2$. But the situation isn't quite so simple, because G isn't the only function of x that can be differentiated to get g . Any real number c can be added to G .





INDEFINITE INTEGRALS

When we differentiate the result, we always get g back again, because the derivative of the constant is always 0. We can think of it like this: $G(x) = x^2$ and therefore:

$$[G(x) + c]' = 2x + 0 = 2x = g(x).$$

That means

$$\int 2x dx = x^2 + c$$

where c can be any real number.





INDEFINITE INTEGRALS

From the above Example 1, we can establish that

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$

where n is a real number with $n \neq -1$.

Example 2.

Evaluate the integral $\int x^4 dx$.

Solution:

Using the above formula, we see that $n = 4$, so that

$$\int x^4 dx = \frac{x^{4+1}}{4+1} + c = \frac{x^5}{5} + c.$$





INDEFINITE INTEGRALS

Example 3.

Evaluate the integral $\int \frac{2}{x^7} dx$.

Solution:

The integral $\int \frac{2}{x^7} dx$ can be written as $\int 2x^{-7} dx$, implying that $n = -7$.

Therefore, we have

$$\int \frac{2}{x^7} dx = \int 2x^{-7} dx = 2 \frac{x^{-7+1}}{-7+1} + c = 2 \frac{x^{-6}}{-6} + c = \frac{-1}{3x^6} + c.$$



INDEFINITE INTEGRALS

Example 4.

Evaluate the integral $\int (11x^{10} - 4x^9 + 1)dx$.

Solution:

The above integral can be split and evaluated in the following manner

$$\begin{aligned}\int (11x^{10} - 4x^9 + 1)dx &= \int 11x^{10}dx - \int 4x^9dx + \int 1dx \\&= 11 \frac{x^{10+1}}{10+1} - 4 \frac{x^{9+1}}{9+1} + 1 \frac{x^{0+1}}{0+1} + C \\&= 11 \frac{x^{11}}{11} - 4 \frac{x^{10}}{10} + 1 \frac{x^1}{1} + C \\&= x^{11} - \frac{2}{5}x^{10} + x + C\end{aligned}$$





TASK

Evaluate the integral $\int (2x^2 - 6x - 3)dx$





DEFINITE INTEGRALS

When a definite integral is evaluated, the result will usually be a number. Definite integrals can be evaluated by applying the following Fundamental Theorem of Calculus.

Theorem.

Let f be a continuous function of x on the interval $[a, b]$. If $F(x)$ is an antiderivative of $f(x)$, then

$$\int_a^b f(x)dx = F(b) - F(a).$$

Sometimes, this is written as

$$\int_a^b f(x)dx = F(x) \Big|_a^b.$$



DEFINITE INTEGRALS

Example 1.

Evaluate the integral $\int_0^3 4x dx$.

Solution:

By the above Fundamental Theorem of Calculus, $f(x) = 4x$
and $F(x) = 2x^2$. Therefore,

$$\int_0^3 4x dx = 2x^2 \Big|_0^3 = 2(3^2 - 0^2) = 18.$$



DEFINITE INTEGRALS

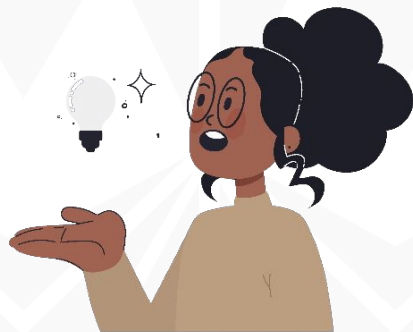
Example 2.

Evaluate the integral $\int_{-1}^5 (3x^2 - 6)dx$.

Solution:

Here, we can have $f(x) = 3x^2$ and $g(x) = -6$, so that $F(x) = x^3$ and $G(x) = -6x$. Therefore,

$$\begin{aligned}\int_{-1}^5 (3x^2 - 6)dx &= 3 \int_{-1}^5 x^2 dx - 6 \int_{-1}^5 x dx \\ &= x^3 \Big|_{-1}^5 - 6x \Big|_{-1}^5 \\ &= (5^3 - (-1)^3) - 6(5 - (-1)) \\ &= 126 - 36 = 90\end{aligned}$$



DEFINITE INTEGRALS

Example 3.

Evaluate the integral $\int_2^3 \frac{1}{x^2} dx$.

Solution:

From the given integral, we have $f(x) = \frac{1}{x^2} = x^{-2}$ so that

$F(x) = \frac{-1}{x}$. Therefore,

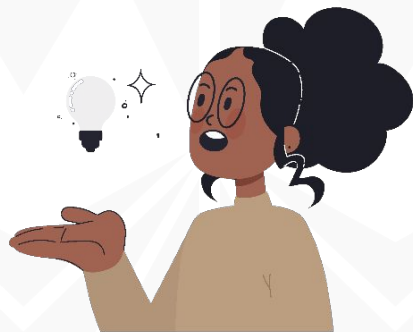
$$\int_2^3 \frac{1}{x^2} dx = \left. \frac{-1}{x} \right|_2^3 = -\left(\frac{1}{3} - \frac{1}{2}\right) = \frac{1}{6}.$$



TASK

Evaluate the integral $\int_{-1}^2 (7x - 5)dx$.





SUMMARY

An indefinite integral is of the form

$$\int f(x)dx = F(x) + c$$

where c is the constant of integration. Indefinite integral is evaluated using the formula

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$

where n is a real number with $n \neq -1$.

A definite integral is evaluated using the Fundamental Theorem of Calculus. It results as

$$\int_a^b f(x)dx = F(b) - F(a).$$



FURTHER READING

" Calculus Know-It-ALL " by Stan Gibilisco.

" Theory and Problems of ADVANCED CALCULUS" by Robert Wrede, and Murray R. Spiegel.

"Advanced Engineering Mathematics" by H. K. Dass.

"Engineering Mathematics" by K. A. Stroud.



THANK
YOU